

The effect of antisialogogues in dentistry

A systematic review with a focus on bond failure in orthodontics

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Successful bonding of orthodontic brackets is a necessary part of orthodontic treatment. Bonding failures can be caused by patient-related factors such as an inability to open the mouth properly, too much saliva and insufficient swallowing. Successful bonding, however, is influenced primarily by operator factors. A significant factor in unsuccessful bonding is moisture contamination, especially with oral fluid. Adequate moisture control is crucial for many dental procedures, from endodontic and restorative treatments to orthodontics. Whereas teeth can be isolated during endodontic and restorative treatment, in orthodontic bonding a larger surface area needs to remain dry and free from saliva contamination. For example, brackets have to be placed on most of the teeth in a dental arch. Moisture contamination is especially difficult to avoid in patients who produce a lot of saliva or do not swallow it in a timely manner. The materials used for bonding are mainly those that will not bond (or that bond insufficiently) when the etched surface is wet. Even though there are hydrophilic materials on the market, the best bond strength still is achieved only when the sur-

ABSTRACT

Background. The authors conducted a literature review to assess whether there is a reduction of salivation with the use of antisialogogues, whether the use of antisialogogues reduces the chair time needed for dental procedures and whether the use of antisialogogues reduces bond failure in orthodontics.

Methods. The authors conducted a search for original articles published from 1950 to April 2010 by using the following databases: Cochrane Collaboration, PubMed, Scopus, EMBASE and ISI Web of Knowledge. They included in their review only human studies in which antisialogogues were used. They validated methodological quality and evidence grade.

Results. Twenty-six studies met the inclusion criteria. Twenty-five of these studies were related to the effect of antisialogogues on salivation, and one study to bond failure. The authors found that there is evidence that antisialogogues work, inconclusive evidence that they reduce bond failure, and no evidence that they reduce chair time for dental procedures.

Clinical Implications. Taking into account the systemic effects of antisialogogues, which exceed the time needed for bracket bonding, the use of antisialogogues for dental procedures in general is questionable.

Key Words. Antisialogogues; anticholinergics; dental bonding; bond failure; orthodontic appliances; dentistry.

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TABLE 1

Indications and effects of parasympatholytic agents with an antisialogogue action.*

DRUG TYPE	DOSE FOR SALIVARY REDUCTION (ADULT DOSE†)	THERAPEUTIC USES	DRUG PROPERTIES	CONTRAINDICATIONS	INTERACTIONS
Atropine	0.4-1.6 milligrams	Parkinson disease; antidote for rapid mushroom poisoning and anticholinesterase intoxication; control of first-degree heart block; ophthalmology: mydriasis and cycloplegia	Blocking/inhibiting acetylcholine action; effects are dose dependent: low dose—depresses salivary, lachrymal, bronchial and sweat secretion, bradycardia; larger dose—dilatation of pupils, photophobia, tachycardia, flushing skin, reduction in tone and mobility of gastrointestinal tract and urinary retention Sedative effect‡	Glaucoma, prostate hypertrophy, myasthenia gravis, obstructive disease of gastrointestinal tract, asthma, allergy to the drug and possibly pregnancy	Antihistamines, tricyclic antidepressants, monoamine oxidase inhibitors and phenothiazine tranquilizers
Scopolamine	0.3-0.6 mg	Sedation and amnesia; motion sickness			
Hyoscyamine	0.125-0.75 mg	Control of bradycardia; reduction of salivation and the secretion of gastric acid during general anesthesia; antidote for rapid mushroom poisoning and anticholinesterase intoxication			
Methantheline	50-100 mg	Peptic ulcers			
Propantheline	15-30 mg	Peptic ulcers			
Glycopyrrolate	1-2 mg	Control of bradycardia; reduction of salivation and the secretion of gastric acid during general anesthesia			

* Sources: Ponduri and colleagues,⁸ Rinchuse and colleagues,⁹ Rinchuse and Rinchuse,¹⁰ Sweetman and Martindale,¹¹ Arzneimittel-kompodium der Schweiz,¹² Sapkos¹³ and Yagiela.^{14,15}

† Pediatric dosage is lower (per kilogram).

‡ Scopolamine.

face of the tooth is dry during bonding.¹⁻⁵

Reductions in the time needed for dental procedures, bonding appliances and procedures performed to maintain a dry working area can make the bonding procedure less cumbersome for the dental practitioner and the patient. Cotton rolls, saliva ejectors, soft-tissue and tongue retractors, and high vacuum suction can be used to help keep the operating field as dry as possible when bonding brackets. However, if a dry operation field is required to bond brackets successfully and to decrease the chair time for bonding, reducing or even stopping salivary flow may be an option. There are several ways to block or reduce salivary flow, including the use of botulinum toxin and the injection and rerouting of the submandibular ducts. These techniques, however, primarily are long-lasting treatments for patients who experi-

ence uncontrollable drooling.^{6,7} For a short, temporary reduction of salivary flow, the remaining options are either prescribing an antisialogogue (a drug that reduces, slows or prevents the flow of saliva) or temporarily blocking the main excretory ducts (for example, with cotton rolls). Antisialogogues have been used in dentistry for many years to reduce salivary flow.⁸ They usually are administered one hour before bonding takes place or a submucosal injection is administered. The most common antisialogogues are antimuscarinic and anticholinergic agents. Such agents have an effect on the central nervous system, but also on

ABBREVIATION KEY. ADA: American Dental Association. GCF: Gingival crevicular fluid. IM: Intramuscular. IV: Intravenous. RCT: Randomized controlled trial.

TABLE 2

Methodological quality grading criteria.*	
GRADE (VALUE OF EVIDENCE)	CRITERIA
A (High)	All criteria should be met: ■ randomized controlled trial or prospective study with a well-defined control group ■ defined diagnosis and end points ■ diagnostic reliability tests and reproducibility tests described ■ blinded outcome measurements
B (Moderate)	All criteria should be met (if not, grade C): ■ cohort study or retrospective case series with a defined control or reference group ■ defined diagnosis and end points ■ diagnostic reliability tests and reproducibility tests described
C (Low)	One or more of the conditions below should be met: ■ large attrition of the sample ■ unclear diagnosis and end points ■ poorly defined patient material

* Source: Bondemark and colleagues.¹⁶

the respiratory, gastrointestinal and cardiovascular systems.⁹⁻¹²

Indications for medical use of parasympatholytic agents with an antisialogogue action are the symptomatic relief of gastrointestinal disorders, treatment of mydriasis (dilation of pupils) and cycloplegia (paralysis of the ciliary muscle of the eye, resulting in loss of adaptation of the pupil). In addition, atropine sometimes is used to dry bronchial and salivary secretion during intubation and upper airway surgery and for reversal of excessive brachycardia.^{11,13} The anticholinergic drugs (the most commonly used type of antisialogogues) that have been recognized by the American Dental Association (ADA) for salivation control and are used in dentistry for that purpose are atropine, hyoscyamine, scopolamine, glycopyrrolate and propantheline,¹⁴ as well as methantheline.¹⁵ However, controlling salivation during dental procedures is not an officially accepted indication for these drugs.¹⁴ Indications and effects of these antisialogogues are shown in Table 1.⁸⁻¹⁵ All of these antisialogogues affect the parasympathetic nervous system, but they have slightly different working mechanisms. Considering the reduction of salivary secretion, which generally is considered an inconvenient side effect of the drugs when used in general medicine, this reduction of salivary flow could be helpful when bonding brackets.

We conducted this systematic review to assess whether there is a reduction of salivation when the antimuscarinic/anticholinergic agents known

as antisialogogues are used, whether the use of antisialogogues reduces the chair time needed for dental procedures, and whether the use of antisialogogues reduces the failure rate of bonded orthodontic brackets.

MATERIALS AND METHODS

Search strategy. To identify publications, we conducted a literature search of articles published from 1950 to April 2010 by using the following databases: ISI Web of Knowledge, the Cochrane Collab-

oration, PubMed, Scopus and EMBASE. There were no language limitations.

We conducted the literature search in each database using the following concatenated search terms: bond OR bonding OR Dental Bonding (MeSH) OR orthodontics (MeSH) OR orthodontic* OR dentistry OR dental care (MeSH) AND (saliva (MeSH) OR salivation (MeSH) OR antisialogogue OR anti-salivation OR salivary OR hyposaliv* OR xerostomi*) AND (banthine OR atropine (MeSH) OR atropine* OR antimuscarinics OR anticholinergics (MeSH)).

For question one (whether there is a reduction of salivation with the antimuscarinic agents mentioned above), we conducted an additional search using the following strategy: (saliva (MeSH) OR salivation (MeSH) OR antisialogogue OR anti-salivation OR salivary OR hyposaliv* OR xerostomi*) AND (banthine OR atropine (MeSH) OR atropine* OR antimuscarinics OR anticholinergics (MeSH) AND humans (MeSH)) AND saliva/drug effects (MeSH). A senior librarian who specialized in health sciences helped us develop the lists of terms and select the databases.

Three authors (A.M.K.-J., A.V., M.A.R.K.) printed out and scored abstracts according to the following inclusion criteria:

- use of drugs that have been recognized by the ADA^{14,15}: atropine, hyoscyamine, scopolamine, glycopyrrolate, methantheline and propantheline;
- human studies;
- effect on salivation;
- use in dentistry for question 2 (whether use of antisialogogues reduces the chair time needed for

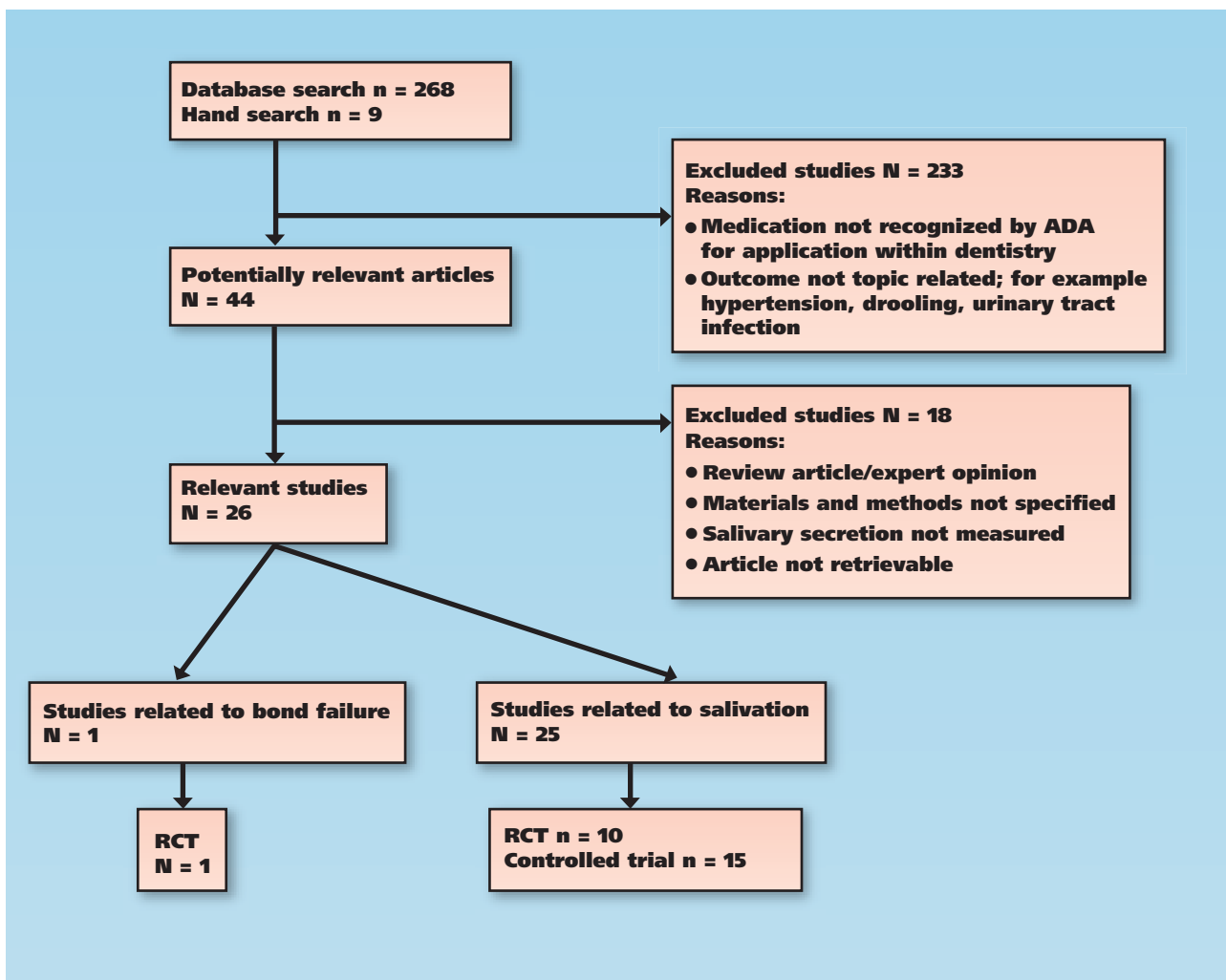


Figure. Quality of Reporting Meta-analyses flow chart of the literature selection process. ADA: American Dental Association. RCT: Randomized controlled trial.

dental procedures) and question 3 (whether it reduces the failure rate of the bonded orthodontic brackets).

We used the broader inclusion criteria term “dentistry” in addition to “orthodontics,” because antisialogogues occasionally are used for the same purpose with other dental procedures.

We excluded reviews and letters, and scored the abstracts as included, excluded or unclear after reviewing the abstract only. We clarified differences among ourselves by means of reaching a consensus. Then we retrieved the full texts of the included articles and articles with unclear abstracts and searched the reference lists for additional articles. We also searched the reference lists of review articles for additional articles. We included and scored any of these additional articles that we thought might be of interest.

Grading the methodological quality of the studies and level of evidence.

We evaluated the studies according to a scoring system developed by the Swedish Council on Technology Assessment in Health Care,¹⁶ which was based on the criteria for assessing study quality from the Centre for Reviews and Disseminations in York, England.¹⁷ The quality grading criteria are listed in Table 2.¹⁶ We judged the final level of evidence for the conclusion of each study according to the following scale.¹⁸

- Strong scientific support (evidence grade 1). Conclusion was based on at least two studies with quality grade A evidence. Studies with opposing conclusions may lower the evidence grade.
- Moderately strong scientific support (evidence grade 2). Conclusion based on one study with strong evidence (quality grade A) and at least two

TABLE 3

Overview of included studies on the effect of antisialogogues on salivation, according to the level of evidence (A, B or C).

AUTHOR (YEAR OF PUBLICATION)	N	DRUG	DOSE	EXPERIMENTAL PERIOD	EFFECT ON SALIVATION
Kazen and Dille¹⁹ (1963)	60	Atropine	0.4, 0.6, 0.8 milligram tablet, capsule or solution	Every 15 minutes up until 3 hours	Saliva flow related to dose
Lönnnerholm and Widerlöv²⁰ (1974)	72	Atropine sulfate	0.25 mg, 0.4 mg, 0.75 mg, 1.5 mg	Baseline, 15, 30, 45 and 60 minutes	Dose-dependent salivary depression
		Atropine methylatropine	0.08 mg, 0.13 mg, 0.25 mg		Dose-dependent salivary depression
Markkanen and Pihlajamäki²¹ (1987)	18	Scopolamine hydrobromide	20 micrograms/kilogram body weight	30 minutes before, 0, 20, 40, 60, 90, 120, 150, 180, 240 and 300 minutes	Reduction of nonstimulated and stimulated salivation, after 5 hours at premedication level
	10	Scopolamine hydrobromide	20 µg/kg body weight	0, 45 and 120 minutes	Reduction of salivation
Wolff and Kleinberg²² (1999)	10	Glycopyrrolate	0.3 mg IM*	Baseline, 0, 15, 60, 105 and 150 minutes	Reduction of salivation, reduction of mucosal wetness, no reduction of gingival crevicular fluid
Brandt and Colleagues²³ (1981)	10	Atropine sulfate	0.4 mg/cubic centimeter submucosal injection	30 minutes preinjection, 5, 20, 35, 50, 65, 80, 95 and 110 minutes; 8, 12 and 22 hours	Initial increase, after 20 minutes decrease, still below control 110 minutes, 8 hours salivation above control
Dobkin and Colleagues²⁴ (1958)	5	Oxyphenonium bromide	1 mg IV [§]	Baseline, 10-, 20- and 30-minute intervals in week between drugs	95% saliva reduction
		Methantheline bromide	10 mg IV		96% saliva reduction
		Promethazine	25 mg IV		93% saliva reduction
		Chlorpromazine	20 mg IV		33% saliva reduction
		Atropine	0.4 mg IV		83% saliva reduction
		Hexocyclium	2 mg IV		97% saliva reduction
Herxheimer²⁵ (1958)	12	Atropine sulfate	0.5 mg/70 kg, 1.0 mg/70 kg, 2.0 mg/70 kg	Patient received two to four different doses, three days between each experiment; baseline observations, time line not clearly mentioned; every hour until 6 hours after	Decreased salivation
		Methantheline bromide	7.0 mg/70 kg, 14.0 mg/70 kg, 28 mg/70 kg, 56 mg/70 kg		
		Propantheline bromide	2.1 mg/70 kg, 4.2 mg/70 kg, 8.4 mg/70 kg, 16.8 mg/70 kg		
		Oxyphenonium bromide	0.5 mg/70 kg, 1.0 mg/70 kg, 2 mg/70 kg, 4 mg/70 kg		
		Hyoscine methylbromide	0.125 mg/70 kg, 0.25 mg/70 kg, 0.5 mg/70 kg, 1 mg/70 kg		
		Hyoscine hydrobromide	0.2 mg/70 kg, 0.4 mg/70 kg, 0.8 mg/70 kg		

* IM: Intramuscular.

† Patient was own control: Each participant had salivation measurements (and, if indicated, other measurements) taken before taking any medication.

§ IV: Intravenous.

¶ Patient was own comparison: Each participant took all drugs with a wash-out period in between.

TABLE 3 (CONTINUED)

REPORTED SIDE EFFECTS	CONTROL	QUALITY GRADE
Not reported	Placebo	B
Tachycardia with two highest doses, highest dose increases diastolic blood pressure	Placebo	B
Bradycardia with lower doses, tachycardia with highest dose		
Decreased heart rate	Placebo	B
Subjective sedation, drowsiness, and blurred vision		
Not reported	Patient was own control [†]	B
Initial bradycardia (n = 4), tachycardia 2 hours after (n = 9)	Patient was own control	C
Tachycardia	Baseline and patient was own comparison [¶]	C
Tachycardia		
Moderately drowsy		
Drowsiness		
Not reported		
Not reported		
Initial tachycardia, increased diameter of pupil, decreased accommodation of pupil	Patient was own control and comparison	C
(continued on following page)		

studies with moderately strong evidence (quality grade B). Studies with opposing conclusions may lower the evidence grade.

■ Limited scientific support (evidence grade 3).

Conclusion based on at least two studies with moderately strong evidence (quality grade B). If studies contradicting the conclusion exist, the scientific basis is judged as insufficient or contradictory.

■ Inconclusive scientific support (evidence grade 4). If studies meeting the evidence criteria are lacking, the scientific basis for the conclusion is considered insufficient.

RESULTS

The Quality of Reporting Meta-analyses flow chart provides an overview of our selection process (Figure). When we searched the databases for studies regarding question 1 about the antisialogogue effects of the medications, we found 124 studies in Scopus, 64 in PubMed, 13 in ISI Web of Knowledge and nine in EMBASE. We found no additional studies in the Cochrane Collaboration database. After we excluded double hits, the search yielded 183 studies. When we searched the databases for studies regarding questions 2 and 3 about antisialogogues' impact on chair time and bonding surfaces versus failure, we found 85 in PubMed, 15 in Scopus, 14 in ISI Web of Knowledge and nine in EMBASE. We found no additional studies in the Cochrane Collaboration database. All of the articles we found in databases other than PubMed we also found in PubMed. In addition, we found nine through hand searching. Forty-four articles met our inclusion criteria. We considered 26 articles to be relevant for this review. One study (a randomized controlled trial [RCT]) was related to bond failure⁸; 25 studies were related to the effect on salivation,¹⁹⁻⁴³ and of these 25 studies, 10 were RCTs.^{19,21,32,34-40}

We graded the quality of the 26 selected articles according to Bondemark and colleagues.¹⁶ The results of these studies follow.

Reduction of salivary flow. Studies regarding the reduction of salivary flow are listed in Table 3¹⁹⁻⁴³ according to their evidence grade. All of the placebo-controlled studies show a reduction of salivation. According to the evidence grading,¹⁸ there is strong scientific support (evidence grade 1) that atropine, hyoscyamine and scopolamine reduce salivary flow from 20 to 300 minutes after the patient took the medication. There is moderately strong scientific support (evi-

TABLE 3

AUTHOR (YEAR OF PUBLICATION)	N	DRUG	DOSE	EXPERIMENTAL PERIOD	EFFECT ON SALIVATION
Murrin ²⁶ (1973)	9	Atropine	7 µg/kg IM or 7, 14, 28 µg/kg orally	Not clear	Salivary depression
Wyant and Dobkin ²⁷ (1957)	11	Atropine	0.2 mg	Carbachal-epinephrine as saliva stimulant; measurements at 0, 10, 20 and 30 minutes before drug administration and every 5 minutes after drug administration	51% reduced saliva secretion
		Scopolamine	0.2 mg		87% reduced saliva secretion
		1-hyoscyamine	0.2 mg		97% reduced saliva secretion
	5	Atropine sulfate	0.4 mg IV	10, 20 and 30 minutes before drug administration and 0, 10, 20 and 30 minutes after drug administration	Reduced saliva secretion
Wyant and Dobkin ²⁸ (1958)	4	Atropine	0.6 mg	No baseline, 10-30 minutes (only stimulated saliva)	96% reduced saliva secretion
		1-hyscyamine	0.3 mg		99.5% reduced saliva secretion
		1-hyoscine	0.2 mg		97% reduced saliva secretion
		Methantheline	5.0 mg		95% reduced saliva secretion
		Hexocyclium	1.5 mg		98.5% reduced saliva secretion
		Oxyphenonium	0.5 mg		99.5% reduced saliva secretion
	5	Atropine	0.6 mg	Every 5 minutes for 1 hour; blood pressure, pulse rate and other impressions	Not measured
		1-hyscyamine	0.3 mg		
		1-hyoscine	0.2 mg		
		Methantheline	5.0 mg		
		Hexocyclium	1.5 mg		
		Oxyphenonium	0.5 mg		
Ishijima and Colleagues ²⁹ (2004)	10	Atropine	0.5 mg	10 minutes before and 10 minutes after medication	Reduction of saliva secretion and masticatory function
Mirakhur and Dundee ³⁰ (1980)	6	Atropine	0.5 mg, 1 mg, 2 mg IM	Baseline, 6 hours observation	Dose-dependent reduction of saliva secretion More prolonged saliva reduction (5.6 times potency of atropine) with glycopyrrolate
		Glycopyrrolate	0.1 mg, 0.2 mg, 0.4 mg IM		
Wyant and Kao ³¹ (1974)	12	Glycopyrrolate	0.1, 0.2 mg	5, 10, 20, 30 minutes, then 2, 3, 4, 5, 7 and 10 hours	Dose-dependent reduction of saliva secretion Prolonged effect with glycopyrrolate
		Atropine	0.4 mg		
Brion and Colleagues ³² (1988)	8	Terfenadine	60 mg	Baseline, 1, 3 and 7 hours after medication	None
		Mequitazine	5 mg		None
		Atropine	1 mg		Atropine produced drop in saliva secretion
Kemmer and Malfertheiner ³³ (1985)	8	Atropine	0.5 mg IV	Before and three times after medication, exact time line not clear	Reduction in salivary secretion
Volz-Zang and Colleagues ³⁴ (1995)	7	Atropine	0.03 mg/kg oral 0.02 mg/kg IM	0, 30, 60, 90, 120, 150, 180 and 210 minutes	Oral: 84.3% reduced saliva secretion maximal at 105 minutes IM: 87.5% reduced secretion maximal at 30 minutes
Rashid and Bateman ³⁵ (1990)	14	Atropine	0.3 mg IV 0.6 mg IV	15, 30, 45, 60, 75 and 90 minutes	Reduced salivary secretion; effect greater in elderly people

TABLE 3 (CONTINUED)

REPORTED SIDE EFFECTS	CONTROL	QUALITY GRADE
28 µm orally showed increased heart rate	Patient was own control	C
Not reported	Patient was own control	C
Pulse rate increased	No baseline, patient was own comparison	C
Pulse rate increased		
Pulse rate decreased		
Pulse rate increased		
Pulse rate decreased		
Pulse rate increased		
Bradycardia with 1-hyoscine, hexocycium Tachycardia with methantheline bromide, oxyphenonium Elevated systolic blood pressure with methantheline and 1-hyoscyamine	Patient was own comparison, no baseline	C
Not reported	Patient was own control	C
Atropine: bradycardia, then tachycardia and change in pupil size Glycopyrrolate: bradycardia	Patient was own control and comparison	C
Not reported	Placebo; patient was own control	C
Bradycardia with atropine in first hour	Placebo; patient was own control	A
Not reported	Patient was own control	C
Oral: decreased heart rate and then increased; IM: immediately increased heart rate	Placebo; patient was own comparison	B
Increased heart rate	Placebo; patient was own comparison	A

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dence grade 2) that glycopyrrolate, methantheline and propantheline reduce salivary flow. The effect seems to be dependent on dose and distribution method. There are, however, only a few studies that examine the effect of dose and distribution method. Regarding the reduction of gingival crevicular fluid (GCF), there is inconclusive scientific support (evidence grade 4) to demonstrate a reduction in flow rate from baseline with the use of glycopyrrolate²²; there also is no evidence that the use of any other antisialogogue reduces the production of GCF.

Reduction of bond failure. The scientific support for the effect of antisialogogue use on bond failure is inconclusive (evidence grade 4), as we identified only one study in which its use was investigated. In a split-mouth design study of 51 patients, Ponduri and colleagues⁸ compared the value of atropine and no atropine on bonding. The bonding was conducted in two sessions (left or right part of the dentition). One hour before one of the sessions, the patients were asked to take a tablet containing 0.6 milligrams of atropine. The researchers did not know which sides of the patients' mouths were bonded when the atropine was taken and could not observe significant differences between bond failure on sides associated with or without atropine use. Unfortunately, the study did not measure patients' compliance but accepted the patients' self-reports. It is unclear whether patient self-reporting has influenced the results.

Reduction of chair time. None of the studies addressed whether chair time was reduced after the use of an antisialogogue.

DISCUSSION

The use of antisialogogues in dentistry is not a new phenomenon and has been described in several review articles.^{5,9,13,14} These reviews all indicated that antisialogogues were useful in reducing salivary flow to facilitate dental treatment, even though they have no officially accepted indication for this type of use in dentistry.¹⁴ Although most literature dates back more than 20 years, the use of antisialogogues seems to have revived somewhat in the last few years. The results of placebo-controlled studies have shown that antisialogogues reduce salivation (Table 3).¹⁹⁻⁴³ However, reduction of chair time and improvement of restorations (for example, increase in integrity, longevity and bonding success) have not been tested. Therefore, the efficiency and efficacy of the antisalivation effects of scopolamine, atropine,

TABLE 3

AUTHOR (YEAR OF PUBLICATION)	N	DRUG	DOSE	EXPERIMENTAL PERIOD	EFFECT ON SALIVATION
Lähteenmäki and Colleagues³⁶ (2000)	12	Glycopyrrolate	0.004 mg/kg IV	0, 1, 3, 6, 12, 24 and 48 hours	Salivary flow rate was reduced for 12 hours
Wilson and Colleagues³⁷ (1984)	10	Methantheline bromide	50 mg oral	Before, 30, 60, 90 and 120 minutes after medication	Both caused reduced salivary secretion; dose-dependent response with clonidine hydrochloride
		Clonidine hydrochloride	0.1 mg, 0.2 mg		
Arneberg and Colleagues³⁸ (1989)	14	Scopolamine	1.5 mg transcutaneous	Before and 1, 2, 3 and 4 days after	Reduced salivary secretion, unstimulated secretion more than stimulated secretion
Litta-Modignani and Colleagues³⁹ (1977)	8	Scopolamine	2.5 mg, 5 mg IV	30, 15 minutes before and 2, 8, 32 and 64 minutes after medication	Reduction in salivary secretion within 2 minutes, baseline level again at 32 minutes
		Hyoscine	10 mg, 20 mg IV		
Ekenved and Colleagues⁴⁰ (1977)	9	Propantheline	30 mg oral	Every hour for 10 hours	Under fasting conditions, reduced secretion, but not when given with food
		I-hyoscyamine	0.8 mg slow release tablet 0.8 mg rapidly disintegrating tablet		Under fasting conditions, reduced secretion Slow release: slower onset of saliva reduction, but effect sustained more
Mirakhur⁴¹ (1978)	6	Atropine	0.5 mg, 1 mg, 2 mg oral or IM	30 minutes before 30, 60 minutes, then 2, 3, 4, 5 and 6 hours	Dose-dependent reduction of salivary secretion and more pronounced with IM Effect more pronounced with hyoscine
		Hyoscine	0.25 mg, 0.5 mg, 1 mg oral or IM		
Grundhofer and Gibaldi⁴² (1977)	4	Propantheline	15 mg and 30 mg	30 minutes before and 0, 30, 60, 90, 120, 150, 180, 210, 240, 270, 300 and 360 minutes	Propantheline showed a greater reduction in salivary secretion, but not all data clear
		Hexocyclium	25 mg and 75 mg		
		Isopropamide	5 mg		
Möller and Rosén⁴³ (1968)	16	Atropine	2 mg, 4 mg oral; 0.33, 1 and 3 mg IM	30 and 15 minutes before; 1, 2, 3, 4, 5, 6, 7 and 8 hours after medication	Dose-dependent reduction in salivary secretion for all medication
		Propantheline	30, 60, 120 mg oral; 3 and 10 mg IM		
		Methylscopolamine	4, 8, 16 mg oral; 0.16 and 0.5 mg IM		
		Butylscopolamine	240, 480 mg oral; 10 and 30 mg IM		

methantheline, propantheline and glycopyrrolate for dental procedures, and for orthodontic use in particular, remain unclear. Furthermore, the only RCT conducted on the effect of atropine on bond failure failed to show a significant effect.⁸ The authors of the study, however, did not assess chair time when bonding and did not assess whether the presence of a dry field reduced the working time. Another limitation of the study was that compliance was not well controlled, because

the patients had to take atropine before visiting the orthodontist. Additionally, no distinction was made between patients with normal salivation and those with hypersalivation so that researchers could observe whether this distinction made a difference in bonding and whether there could be an indication for people with hypersalivation to use atropine. More studies are needed to support these results⁸ to conclude whether antisialogogue use has a place in dentistry.

TABLE 3 (CONTINUED)

REPORTED SIDE EFFECTS	CONTROL	QUALITY GRADE
Not reported	Placebo	A
Not reported	Placebo; patient was own control	A
Not reported	Placebo; patient was own comparison	A
Dose-related increase in heart rate for 30 minutes; adaptation of pupil reduced	Patient was own comparison	A
Not reported	Placebo; patient was own control	A
Dose-dependent increase in heart rate more pronounced with IM Pupil dilation	Patient was own control and comparison	C
Bradycardia Pupil dilation		
Not reported	Patient was own control	C
Tachycardia; initial bradycardia with propantheline; IM: more pronounced and earlier onset	Variety of controls, not well defined	C

Although the use of antisialogogues seems to have been adopted by some orthodontists as a useful aid when bonding brackets, there is no evidence that antisialogogues facilitate the bonding procedure or reduce the bonding failure rate. It is possible that the bonding procedure used is the determining factor, regardless of whether or not an antisialogogue is used. The results of one study showed that the reduction of flow rate of GCF was negligible after glycopyrrolate use²²;

however, it has not been studied with the other drugs used. Therefore, the questions remain whether GCF does affect the bonding of brackets and whether using an antisialogogue will reduce the secretion of GCF and achieve reductions in chair time and bond failure. These topics need further study.

The adverse effects of several antisialogogues were not properly studied in any of the studies we included in our review; for example, ocular effects and effects on heart rate and blood pressure were not always studied. This is a clinically relevant omission, as pharmaceutical resources^{11,12} warn about atropine use in particular, owing to its unpredictable individual response. Moreover, adverse effects and intoxication effects from antisialogogue use are not dose dependent. These adverse effects might be even more important when patients also take other medications. The same unpredictable individual response has been reported for scopolamine.⁴⁴ Furthermore, when a patient is undergoing a bonding procedure only, the effect of the given drugs might last longer than the patient's visit and might affect the patient's physical status or health for the rest of the day. Carter⁴⁵ warned against patients' being active (for example, participating in sporting activities) after taking an antisialogogue because hyperthermia and problems with contact lenses due to dry eyes might occur. Therefore, how long the antisialogogue continues to work when these patients return back to school, work or home remains unknown.

Regarding the willingness of patients to use atropine as part of their orthodontic treatment, the authors of only one questionnaire-based study reported that most parents and patients found it acceptable for patients to take atropine before their brackets were bonded.⁸ These findings, however, may be due in part to the Hawthorne effect, which suggests that participants report favorably to an experimental manipulation simply in response to the fact that they are being studied.⁴⁶

When evaluating the evidence, which consists of the results of only one study that show no difference in bond failure rates,⁸ opinions can differ. Some authors say that the existence of one study that does not demonstrate any difference does not necessarily mean that every method that enables salivation reduction should be abandoned,⁴⁷ while other authors are reluctant to have patients premedicate because of the risk of side effects.¹⁰ Therefore, one may wonder if it is defensible to

give patients a medication without a true medical indication simply to make the bonding procedure easier or to prevent bracket failure, especially when these medications can have lasting effects after the patient has left the office.^{11,21,23}

Considering possible alternatives for medication, one might consider hypnosis or other methods that would allow patients to sleep during the procedure to take advantage of the fact that there is a reduction in salivation during sleep.⁴⁸ These alternatives, however, probably would not reduce chair time and all patients might not sleep with their mouths open, leading to additional procedures that might need to be performed to prevent patients from closing their mouths. Another solution that could be developed would be a device that can easily stop salivation for a certain period without damaging the salivary glands.

CONCLUSION

There is no conclusive scientific evidence to support the use of antisialogogues during dental procedures or to reduce bond failure and chair time. The side effects of using this specific medication for a nonmedical indication have not been examined properly. ■

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- Webster MJ, Nanda RS, Duncanson MG Jr, Khajotia SS, Sinha PK. The effect of saliva on shear bond strengths of hydrophilic bonding systems. *Am J Orthod Dentofacial Orthop* 2001;119(1):54-58.
- Cacciabesta V, Sfondrini MF, De Angelis M, Scribante A, Klersy C. Effect of water and saliva contamination on shear bond strength of brackets bonded with conventional, hydrophilic, and self-etching primers. *Am J Orthod Dentofacial Orthop* 2003;123(6):633-640.
- Reddy L, Marker VA, Ellis E 3rd. Bond strength for orthodontic brackets contaminated by blood: composite versus resin-modified glass ionomer cements. *J Oral Maxillofac Surg* 2003;61(2):206-213.
- Paschos E, Westphal JO, Ilie N, Huth KC, Hickel R, Rudzki-Janson I. Artificial saliva contamination effects on bond strength of self-etching primers. *Angle Orthod* 2008;78(4):716-721.
- Sherman CR, Sherman BR. Atropine sulfate: a current review of a useful agent for controlling salivation during dental procedures. *Gen Dent* 1999;47(1):56-60.
- Reed J, Mans CK, Brietzke SE. Surgical management of drooling: a meta-analysis. *Arch Otolaryngol Head Neck Surg* 2009;135(9):924-931.
- Scully C, Limeres J, Gleeson M, Tomás I, Diz P. Drooling. *J Oral Pathol Med* 2009;38(4):321-327.
- Ponduri S, Turnbull N, Birnie D, Ireland AJ, Sandy JR. Does atropine sulphate improve orthodontic bond survival? A randomized clinical trial. *Am J Orthod Dentofacial Orthop* 2007;132(5):663-670.
- Rinchuse DJ, Rinchuse DJ, Sprecher R. Clinical pharmacology for the orthodontist. *Am J Orthod* 1981;79(3):273-281.
- Rinchuse DI, Rinchuse DI. There are two sides to the coin (Readers' Forum). *Am J Orthod Dentofacial Orthop* 1999;115(3):17A.
- Sweetman SC, Martindale W. Martindale: The Complete Drug Reference. 34th ed. London: Pharmaceutical Press; 2005:475-479, 482-485, 489.
- Arzneimittel-kompodium der Schweiz. "www.compendium.ch/Search.aspx?lang=de". Accessed June 16, 2010.
- Sapkos SW. The use of antisialogogues in periodontal and restorative dentistry. *Int J Periodontics Restorative Dent* 1984;4(4):42-49.
- Yagiela JA. Agents affecting salivation. In: ADA/PDR Guide to Dental Therapeutics. 5th ed. Ciancio SG, ed. Chicago: American Dental Association; 2009:291-304.
- Yagiela JA. Agents affecting salivation. In: ADA Guide to Dental Therapeutics. 2nd ed. Ciancio SG, ed. Chicago: American Dental Association; 2000:198-210.
- Bondemark L, Holm AK, Hansen K, et al. Long-term stability of orthodontic treatment and patient satisfaction: a systematic review. *Angle Orthod* 2007;77(1):181-191.
- Deeks J, Glanville J, Sheldon T. Undertaking Systematic Reviews of Research on Effectiveness: CRD Guidance for Those Carrying Out or Commissioning Reviews. York, England: NHS Centre for Reviews and Dissemination; 2001. CRD report no. 4.
- Mohlin B, Axelsson S, Paulin G, et al. TMD in relation to malocclusion and orthodontic treatment. *Angle Orthod* 2007;77(3):542-548.
- Kazen DH, Dille JM. An evaluation of atropine as an antisialogogue in dentistry. *Oral Surg Oral Med Oral Pathol* 1963;16:919-925.
- Lönnnerholm G, Widerlöv E. Effect of intravenous atropine and methylatropine on heart rate and secretion of saliva in men. *Eur J Clin Pharmacol* 1975;8(3-4):233-240.
- Markkanen YJ, Pihlajamäki K. Oral scopolamine hydrobromide solution as an antisialogogic agent in dentistry. *Oral Surg Oral Med Oral Pathol* 1987;63(4):417-420.
- Wolff MS, Kleinberg I. The effect of ammonium glycopyrrolate (Robinul)-induced xerostomia on oral mucosal wetness and flow of gingival crevicular fluid in humans. *Arch Oral Biol* 1999;44(2):97-102.
- Brandt S, Servoss JM, Persily KB. Atropine sulphate: an effective antisialogogue. *J Clin Orthod* 1981;15(9):629-634.
- Dobkin AB, Wyant GM, Aasheim GM. Antisialogogue drugs in man: comparison of some anticholinergic and sedative antihistamine drugs. *Anaesthesia* 1958;13(1):63-67.
- Herxheimer A. A comparison of some atropine-like drugs in man, with particular reference to end-organ specificity. *Br J Pharmacol Chemother* 1958;13(2):184-192.
- Murrin KR. A study of oral atropine in healthy adult subjects. *Br J Anaesth* 1973;45(5):475-480.
- Wyant GM, Dobkin AB. Antisialogogue drugs in man; comparison of atropine, scopolamine (1-hyoscyne) and 1-hyoscyamine (bellafoline). *Anaesthesia* 1957;12(2):203-214.
- Wyant GM, Dobkin AB. Further studies of antisialogogue drugs in man. *Anaesthesia* 1958;13(2):173-178.
- Ishijima T, Koshino H, Hirai T, Takasaki H. The relationship between salivary secretion rate and masticatory efficiency. *J Oral Rehabil* 2004;31(1):3-6.
- Mirakhur RK, Dundee JW. Comparison of the effects of atropine and glycopyrrolate on various end-organs. *J R Soc Med* 1980;73(10):727-730.
- Wyant GM, Kao E. Glycopyrrolate methobromide, 1: effect in salivary secretion. *Can Anaesth Soc J* 1974;21(2):230-241.
- Brion N, Beaumont D, Advenier C. Evaluation of the antimuscarinic activity of atropine, terfenadine and mequitazine in healthy volunteers. *Br J Clin Pharmacol* 1988;25(1):27-32.
- Kemmer T, Malfertheiner P. Influence of atropine on taste-stimulated parotid secretion. *Res Exp Med (Berl)* 1985;185(6):495-502.
- Volz-Zang C, Walldhäuser T, Schulte B, Palm D. Comparison of the effects of atropine in vivo and ex vivo (radioreceptor assay) after oral and intramuscular administration to man. *Eur J Clin Pharmacol* 1995;49(1-2):45-49.
- Rashid MU, Bateman DN. Effect of intravenous atropine on gastric emptying, paracetamol absorption, salivary flow and heart rate in young and fit elderly volunteers. *Br J Clin Pharmacol* 1990;30(1):25-34.
- Lähtenmäki MT, Salo MS, Tenovu JO, Helminen AV, Vilja PJ, Huupponen RK. The effects of glycopyrrolate on oral mucous host defenses in healthy volunteers. *Anesth Analg* 2000;91(2):467-472.
- Wilson EL Jr, Whitsett LD, Whitsett TL. Effect of methantheline bromide and clonidine hydrochloride on salivary secretion. *J Prosthet Dent* 1984;52(5):663-665.
- Arneberg P, Storhaug K, Sandvik L. Effect of a slow release transcutaneous scopolamine application on salivary flow, pH, buffering action, and salivary levels of *Streptococcus mutans* and lactobacilli. *Scand J Dent Res* 1989;97(5):408-414.
- Litta-Modignani R, Mazzolari M, Barantani E. Relative potency of the atropine-like effects of a new parasympatholytic drug, scopolamine-N-(cyclopropyl methyl) bromide and those of hyoscyne-N-butyl bromide. *Curr Med Res Opin* 1977;5(4):333-340.
- Ekenved G, Magnusson A, Bodemar G, Wälan A. Influence of food on the effect of propantheline and L-hyoscyamine on salivation. *Scand J Gastroenterol* 1977;12(8):963-966.
- Mirakhur RK. Comparative study of the effects of oral and i.m. atropine and hyoscyne in volunteers. *Br J Anaesth* 1978;50(6):591-598.
- Grundhofer B, Gibaldi M. Biopharmaceutic factors that influence effects of anticholinergic drugs: comparison of propantheline, hexocy-

clium, and isopropamide. *J Pharm Sci* 1977;66(10):1433-1435.

43. Möller J, Rosén A. Comparative studies on intramuscular and oral effective doses of some anticholinergic drugs. *Acta Med Scand* 1968;184(3):201-209.

44. Corallo CE, Whitfield A, Wu A. Anticholinergic syndrome following an unintentional overdose of scopolamine. *Ther Clin Risk Manag* 2009;5(5):719-723.

45. Carter RN. Salivary control. *J Clin Orthod* 1981;15(8):562-564.

46. Rinchuse DJ, Sweitzer EM, Rinchuse DJ, Rinchuse DL. Understanding science and evidence-based decision making in orthodontics. *Am J Orthod Dentofacial Orthop* 2005;127(5):618-624.

47. Kapit AL. Does atropine sulphate improve orthodontic bond survival? *Am J Orthod Dentofacial Orthop* 2008;133(2):185.

48. Dawes C. Factors influencing salivary flow rate and composition. In: Edgar WM, O'Mullane DM, eds. *Saliva and Oral Health*. London: British Dental Association; 1996:27-44.